3. Albert

The father of Albert Einstein built power stations in Italy. When Albert was a young boy, the Maxwell equations were only a few decades old – but Italy was entering its industrial revolution, and the turbines and transformers that his father constructed were already based on them. The power of the new physics was obvious.

Albert was a rebel. His parents left him in Germany to attend high school, but he found the German school system too rigid and militaristic; he could not stand the authority of the school and abandoned his studies. He joined his parents in Italy, in Pavia, and spent his time loafing. Later, he went to study in Switzerland, initially failing to get into the Zurich Polytechnic, as he wished. After his university years he could not find a research position and, in order to live with the girl he loved, he found employment in the patent office of Berne.

It wasn't much of a profession for a physics graduate, but it gave Albert time to think, and to work independently. And he did think and work. After all, this is what he had done since his early youth: he would read Euclid's *Elements* and Kant's *Critique of Pure Reason* instead of attending to what he was being taught at school. You don't get to new places by following established tracks.

At the age of twenty-five, Einstein sends to the *Annalen der Physik* three articles. Each was worthy of a Nobel Prize, and more. Each one of the three is a pillar supporting our understanding of the world. I have already spoken of the first article, in which the young Albert calculates the dimensions of atoms and proves, after twenty-three centuries, that the ideas of Democritus were correct: matter is granular.

The second article is the one for which Einstein is most famous – the article in which he introduces the theory of relativity – and it is to the theory of relativity that this chapter is devoted.

In fact, there are two theories of relativity. The envelope sent by the twenty-five-year-old Einstein contained the exposition of the first of these: the theory known today as 'special relativity'. This is an important clarification of the structure of space and time, which I illustrate here before turning to the other, and most important of Einstein's theories: general relativity.

Special relativity is a subtle and conceptually difficult theory. It is more difficult to digest than general relativity. Reader, don't become demoralized if